



Palouse Alliance Community Resource Platform

03/24/2026 - 05/01/2026

Client: Allison Rockwell
05-SP26-FA26-PAHIFC-WEBMOB



Client and Team Information

Project Description

The Palouse Alliance Community Research Platform is a centralized website designed to help Palouse residents find local events, services and resources. It also helps community members post and advertise events for users to attend.

Team Roles

- Vijay Deva (Integration Manager, Team Lead)
- Omar Adil (Frontend Developer)
- Kaeden Peterson (Backend Developer)
- Aryan Puthran (Database Manager, QA Manager)



Sprint Objectives

Main Sprint Goals

- Centralize community resources
- Improve discoverability and accuracy of local services
- Provide a clear and accessible interface usable across all devices
- Support organizations with easier updates and coordinated communication

Rationale and Vision

This sprint focused on creating the foundational steps for the application and improving them so the next few sprints are left for the few non-functional requirements. To do this we focused on creating event creation method, secure database integration, user experience, and admin dashboard, since these are all critical user stories/roles.



Problem Statement

Residents in the Palouse region struggle to find reliable information about local services due to resources spread out across websites, flyers, and social media posts. This makes it hard to know what help exists or whether information is current or not.

Organizations face similar challenges. They must repeatedly share the same details across platforms, which often leads to outdated content, inconsistent communication, and extra maintenance. Without a unified system, both residents and service providers lose time and miss important opportunities.



Scope

This project will deliver a unified, accessible platform that centralizes community resources so residents can find accurate, recent information with ease. It focuses on reducing fragmentation, improving discoverability, and supporting organizations with simpler, most consistent communication tools. The scope includes both the public users who need reliable services and the Alliance members who maintain and share information.

Target Audience

- Residents seeking community health, wellness, and support services
- Individuals looking for housing help, food assistance, mental health resources, or local programs
- Students, families, seniors, and veterans across the Palouse region.
- Alliance organizations that post updates, events, and service information.



Technologies Used

The project is built using a React-based frontend with JSX component, CSS, and JavaScript for interactive behavior. Development runs on Vite for fast and efficient build process.

Navigation is managed through React Router, while local assets such as logos are imported directly into components.

Styling is organized through dedicated CSS files, creating a lightweight and maintainable workflow.

All application data is handled through and SQL-based backend, ensuring structured and reliable storage.



Functional Requirements

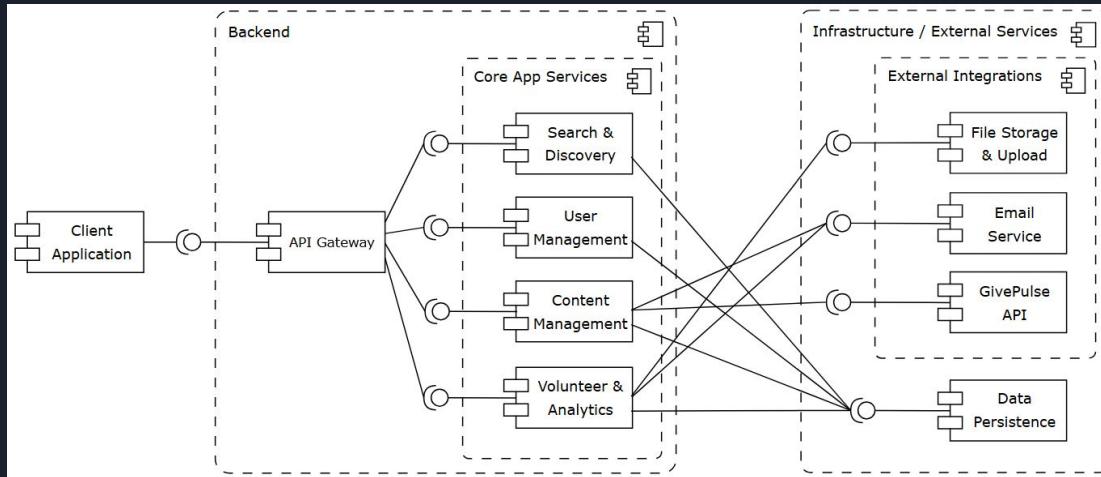
- Search and filtering: Keyword search with category and tag filters for services and events
- Events system: Public event calendar with approved postings, flyers, and meetings that are viewable to users; email notifications for new resources
- User management: Member registration/login with verified accounts; admin approval for users and content
- Content and community: Event/flyer uploads, organization listings, and GivePulse integration
- Information and privacy: Public display of contact info with user-controlled personal data sharing options
- Platform features: Responsive design, branding support, custom domain, and search engine visibility
- Admin tools: Basic analytics (registrations, attendance tracking) for administrators



Non-functional Requirements

- Usability: Interface should be intuitive and accessible; content upload should be as easy as sending an email; should follow accessibility standards
- Scalability: Handles growth in number of users and events without performance degradation
- Reliability: System should be available 99% of the time with minimal downtime
- Maintainability: Modular and documented tech stack; code reviews and version control
- Security and privacy: Secure storage of sensitive data; authentication required; users control visibility of personal info
- Platform compatibility: Accessible across modern browsers and devices (desktop/laptop and mobile)
- Brand consistency: Consistent application of provided brand logos and colors

Architecture of the System



- Client-server, Layered Architecture
- Separate Frontend, Backend, Database, and External Services
- Designed for scalability, maintainability, and security

Core Components:

- Frontend/UI - (React)
- Backend - ([Node.js](#) + API Layer)
 - Logic for authentication, routing, and moderation
 - Database - (Supabase/PostgreSQL)
 - Stores users, events, orgs, tags, etc.
- Admin Dashboard
 - Event approval, user management, analytics
- External Integrations
 - GivePulse, Email Notifications (future sprint)



Sprint 3 Achievements - Vijay Deva

- Implemented Event Calendar System
 - Designed calendar component that enables text, tag, and category filtering
 - Pulls all approved events from database and displays all necessary info through event cards
 - Supports multi-day events and zoom events
- Implemented My Events tab to display users events
 - Pulls user dedicated events from database
 - Enables filtering from approved, pending, and rejected
- Deployed codebase on vercel
 - Created Preview and Production Pipelines for the platform
- Assisted with backend post of “Post Event”

This sprint I wanted to work on the core features of the platform so we can get in shape for deployment. We were able to complete most of the core features we had in mind, so we were able to publish the codebase. We had some issues with the design and layout of the website this sprint, but speaking with the client and doing a full rework of the website to clear some things helped a ton.



Sprint 3 Achievements - Omar Adil

- Implemented the primary home page and refining it to include information provided by our client.
- Refactored the global footer to incorporate organization information, making sure its consistent across all platform routes.
- Conducted testing to identify and resolve logic and UI conflicts in order to improve stability of the application.

Sprint 3 was very smooth because the foundation we built in earlier sprints made it easy to implement these new changes. The only challenges were a few minor bugs found during testing, but I communicated the issue with our database manager to get those fixed quickly. Overall, our team's communication and groundwork allowed us to stay on track and deliver a polished product.



Sprint 3 Achievements - Kaeden Peterson

- Made iterative improvements and refinements to the Post Event feature, enhancing the overall structure and usability of the form across all steps
- Improved user experience with frontend state management for form progression, validation, and data persistence across steps
- Completed integration with the Supabase backend to successfully insert event data into the database while aligning with the required schema
- Refined data handling and formatting to resolve issues with database constraints

Sprint 3 went well overall. The foundation established in earlier sprints aligned well with our goal of completing a fully functional event submission feature. The main challenges involved ensuring frontend data matched backend requirements. Our team's coordination and prior work allowed us to stay on track and polish the feature.



Sprint 3 Achievements - Aryan Puthran

- Implemented a full-featured admin dashboard for managing users and organizations, including registration and password resets.
- Added admin event management with approve/reject workflows and real-time dashboard refetching.
- Improved event UX with clearer date/time display, calendar filtering, multi-day event support, and consistent “My Events” behavior.
- Streamlined post-event submission with a unified date/time picker, URL support and validation, success toast, and redirect.
- Optimized caching and multi-account handling to prevent unnecessary event reloads.
- Various Improvements: mobile fixes, icons, and Vercel deployment preparation.

A goal of mine was to implement admin features in Sprint 3, and I accomplished that. I didn't face any challenges since the database was fully setup to support the admin side of things.



System Demonstration

- Demo Readiness & Completion
 - Feature completion: ~85% of planned features implemented.
 - Testing: layered testing completed for critical flows (unit, integration, E2E, accessibility).
- User Interaction & Alignment
 - Majority of user stories completed (~85%). These stories were defined from client requirements and input.
 - Features simple discovery, easy event posting, and admin moderation aligned with needs.
- Live demo!



Testing

- Testing Methodologies
 - Unit testing (core components & utilities)
 - Integration testing (frontend <-> backend, auth, database)
 - System / end-to-end testing (critical user flows: search, post event, calendar)
 - Performance & load checks (page load, API response times)
 - Accessibility testing (automated audits + manual assistive-tech checks)
- Key Results & Metrics
 - Coverage: unit + integration tests for primary modules (coverage target: 70–90%)
 - Critical flows: search and event posting exercised in E2E tests (pass rate: high for stable flows)
- Testing Challenges & Resolutions
 - Challenge: accessibility edge-cases not caught by automated tools
 - Resolution: manual testing with only keyboard navigation, and color contrast checks



Responsible Use of AI

- AI was not used for any setup or documentation in repository.
- AI was used for minor bug fixes relating to the caching system and styling.
- AI was used for spell check in our team submissions, which allowed for more concise and better communication.
 - All changes were reviewed and approved by a team member before being implemented.
- We didn't depend on AI enough to produce errors or notice limitations.

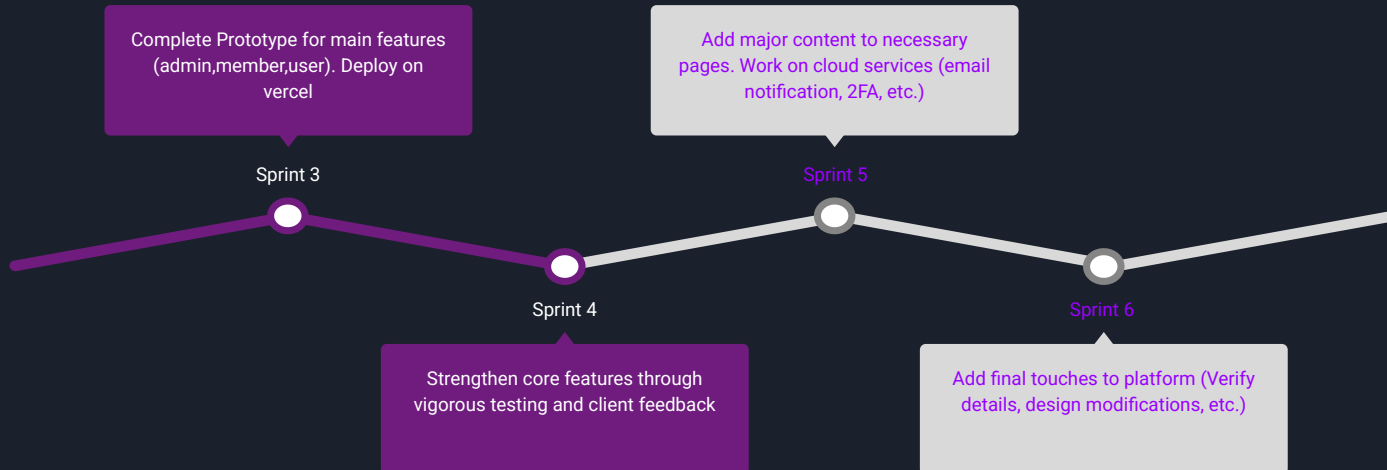
Plan and Deliverables for Next Semester

Planned Functional Requirements:

- [Finalizing Flyer Storage and Retrieval](#)
- [View Flyers](#)
- [Event Retention](#)
- [Analytics and attendance tracking](#)
- [Email Notifications](#)

Planned Non-Functional Requirements

- Brand Consistency
- Maintainability
- Reliability
- Usability





Conclusion

In this sprint, we finalized the core features for the admins, members and users. Users are able to discover events, members are allowed to post events, and admins are allowed to reject/approve events. We have also deployed this platform on vercel which can be seen [here](#).

In our next sprint we plan to strengthen our core features through rigorous testing and client feedback. We then plan to move onto other features like email notifications and 2FA.



Thank you for listening!

Any feedback for our work this semester?